Project 1

# NMAP SCANNING TOOLS

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Network Security Fundamentals

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Introduction

This project demonstrates the use of Nmap to perform network scanning and enumeration on a multiple public targets. The scan types were conducted using Nmap 7.89 through macOS terminal and the Zenmap graphical interface. Various Scanning techniques were used to identify open ports, running services, service versions, operating system information, and firewall behavior.

The objective of this project is to analyze how different Nmap scan types reveal information about remote hosts and to understand how network security mechanism affects scan results.

Target Information

Target 1- www.target.com

Organization: Target Corporation

Scanning Tool: Nmap 7.98

Operating System Used: macOS (Terminal)

Step 1- Host Discovery Scan

Command Used

nmap -sP www.target.com

```
Last login: Mon Feb 16 10:40:05 on ttys000
/Users/rajulama/.zprofile:export:1: not valid in this context: PATH,
rajulama@MacBookPro ~ % nmap -sP www.target.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 10:40 -0500
Nmap scan report for www.target.com (199.232.90.187)
Host is up (0.0052s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.63 seconds
```

(Figure 1: Host Discovery Scan Output)

Purpose

The purpose of this scan was to verify whether the target host is active and reachable before performing port scanning.

Results

The scan confirmed that www.target.com is up with low latency (0.0052 seconds).

Analysis

The response confirms that the host is reachable and ready for further scanning.

Step 2- TCP Ping Scan

Command Used

nmap -sP -PT80 www.target.com

Screen Shot

```
rajulama@MacBookPro ~ % nmap -sP -PT80 www.target.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 13:49 -0500
Nmap scan report for www.target.com (199.232.90.187)
Host is up (0.0048s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.55 seconds
```

(Figure 2: TCP Ping Scan Output)

Purpose

This scan checks if the host is alive by sending a TCP packet to port 80 instead of using normal ICMP ping.

Results

The scan confirmed that the host is up.

Analysis

The target responds to TCP traffic on port 80. This confirms that web services are reachable even if ICMP ping is blocked.

Step 3- TCP Connect Scan

Command Used

nmap -T4 -sT www.target.com

Screenshot

```
rajulama@MacBookPro ~ % nmap -T4 -sT www.target.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 10:42 -0500
Stats: 0:00:17 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 37.95% done; ETC: 10:42 (0:00:28 remaining)
Nmap scan report for www.target.com (151.101.46.187)
Host is up (0.0092s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE
80/tcp   open  http
443/tcp  open  https

Nmap done: 1 IP address (1 host up) scanned in 43.64 seconds
```

(Figure 3: TCP Connect Scan Output)

Purpose

The TCP Connect scan (-sT) was performed to identify open TCP ports on the target system. The -T4 option was used to increase scan speed while maintaining reliability.

Results

The scan identified two open ports:

- Port 80 (HTTP)
- Port 443 (HTTPS)

All other 998 scanned ports were filtered, indicating no response from those ports.

Analysis

The presence of open ports 80 and 443 confirms that the target exposes web services to the public. The remaining filtered ports suggest that a firewall is actively blocking access to non-essential services. This shows that the server only allows web services and block other ports.

Step 4- SYN Stealth Scan (-sS)

Command Used

sudo nmap -sS -Pn www.target.com

Screenshot

```
rajulama@MacBookPro ~ % sudo nmap -sS -Pn www.target.com

Password:
Sorry, try again.
Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 10:45 -0500
Nmap scan report for www.target.com (199.232.90.187)
Host is up (0.017s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 8.74 seconds
```

(Figure 4: SYN Stealth Scan Output)

Purpose

This scan was done to check which ports are open using a SYN scan. This type of scan does not complete the full connection. It is often used to scan in a quiet way. The -Pn option tells Nmap not to check if the host is up first.

Results

The scan showed that port 80 and port 443 are open. All other 998 ports were filtered.

Analysis

Port 80 is used for HTTP.

Port 443 is used for HTTPS.

The filtered ports mean that a firewall is blocking most of the ports. This shows that the website only allows web traffic and blocks other services.

Step 5- UDP Scan

Command Used

sudo nmap -sU www.target.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sU www.target.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 10:47 -0500
Nmap scan report for www.target.com (151.101.46.187)
Host is up (0.0065s latency).
All 1000 scanned ports on www.target.com (151.101.46.187) are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 22.78 seconds
```

(Figure 5: UDP Scan Output)

Purpose

This scan was done to check for open UDP ports on the target.

Results

The scan showed that all 1000 UDP ports are open or filtered, meaning no clear UDP services responded.

Analysis

No UDP services were clearly found open. This likely means the firewall is blocking UDP traffic. The server only allows necessary services.

Step 6- One Stealth Variation

Command Used

sudo nmap -sF www.target.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sF www.target.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 10:51 -0500
Nmap scan report for www.target.com (199.232.90.187)
Host is up (0.0069s latency).
All 1000 scanned ports on www.target.com (199.232.90.187) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.83 seconds
```

(Figure 6: FIN Stealth Scan Output)

Purpose

This scan was done to perform a stealth scan using the FIN flag. The goal is to detect open ports without completing a full connection.

Results

The scan shows that all 1000 TCP ports are in ignored states. It also shows that 1000 ports are filtered.

Analysis

The server did not respond to the FIN scan. This usually means a firewall is filtering the packets. The firewall is blocking unusual scan types like FIN scans. This shows the server has security protection enabled.

Step 7- Xmas Scan

Command Used

sudo nmap -sX www.target.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sX www.target.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 14:00 -0500
Nmap scan report for www.target.com (151.101.46.187)
Host is up (0.0080s latency).
All 1000 scanned ports on www.target.com (151.101.46.187) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 22.45 seconds
```

(Figure 7: Xmas Scan Output)

Purpose

This scan sends packets with FIN, URG, and PUSH flags set. It is used to test firewall filtering behavior.

Results

The scan showed that all 1000 TCP ports are in ignored states. No open ports were detected.

Analysis

The server did not respond to the Xmas scan packets. This indicates that a firewall is filtering unusual scan types. The system is protected against this stealth scanning method.

Step 8- Null Scan

Command Used

sudo nmap -sN www.target.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sN www.target.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 14:10 -0500
Nmap scan report for www.target.com (199.232.90.187)
Host is up (0.0066s latency).
All 1000 scanned ports on www.target.com (199.232.90.187) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 22.53 seconds
```

(Figure 8: Null Scan Output)

Purpose

This scan sends TCP packets with no flags set. It is used to test firewall behavior and filtering rules.

Results

The scan showed that all 1000 TCP ports are in ignored states. No open ports were detected.

Analysis

The target system did not respond to the Null scan packets. This indicates that the firewall is blocking abnormal scan traffic. The system is properly protected against this scanning technique.

Step 9- OS Detection

Command Used

sudo nmap -sS -O www.target.com

Screen shot

```
rajulama@MacBookPro ~ % sudo nmap -sS -O www.target.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 10:53 -0500
Nmap scan report for www.target.com (199.232.90.187)
Host is up (0.0075s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed po
rt
Aggressive OS guesses: Apple iOS 14.0 - 15.6 or tvOS 14.3 - 16.1 (Darwin 20.0.0 - 22.1.0) (87%), Ap
ple macOS 11 (Big Sur) - 13 (Ventura) or iOS 16 (Darwin 20.6.0 - 22.4.0) (87%), Crestron XPanel con
trol system (86%), Android 9 - 10 (Linux 4.9 - 4.14) (86%), Android 8 - 9 (Linux 3.18 - 4.4) (86%),
 OpenWrt 0.9 - 7.09 (Linux 2.4.30 - 2.4.34) (86%), OpenWrt White Russian 0.9 (Linux 2.4.30) (86%),
OpenWrt Kamikaze 7.09 (Linux 2.6.22) (86%), OpenBSD 4.3 (85%), Linux 2.6.32 - 3.10 (85%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.60 seconds
```

(Figure 9: OS Detection Scan Output)

Purpose

This scan was done to detect the operating system of the target server.

Results

Port 80 is open for HTTP. Port 443 is open for HTTPS.

Nmap tried to guess the operating system. It gave several possible matches such as Apple iOS, macOS, Android, Linux, and OpenBSD.

It also shows; “No exact OS matches for host."

Analysis

The scan could not determine the exact operating system. This usually happens when a firewall blocks some responses. It may also happen because the server is behind a content delivery network. The result shows that OS fingerprinting was attempted, but the exact system could not be confirmed.

Step 10- Comprehensive Scan

Command Used

sudo nmap -sC -sV www.target.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sC -sV www.target.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 10:56 -0500
Nmap scan report for www.target.com (151.101.46.187)
Host is up (0.020s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE    VERSION
80/tcp  open  http-proxy Varnish
|_http-title: Did not follow redirect to https://www.target.com/
443/tcp open  ssl/https
| ssl-cert: Subject: commonName=sites.target.com/organizationName=Target Corporation/stateOrProvinc
eName=Minnesota/countryName=US
| Subject Alternative Name: DNS:sites.target.com, DNS:affiliate.target.com, DNS:apollo-metrics.targ
et.com, DNS:assethub.partnersonline.com, DNS:assethub.target.com, DNS:bex.partnersonline.com, DNS:b
ex.target.com, DNS:cartster.target.com, DNS:cartwheel.target.com, DNS:cartwheelws-secure.target.com
, DNS:circle.target.com, DNS:connect.roundel.com, DNS:corporate.target.com, DNS:developer.target.co
m, DNS:doppler.partnersonline.com, DNS:elevate.target.com, DNS:extgargantua.partnersonline.com, DNS
:factorial.partnersonline.com, DNS:finds.target.com, DNS:greenfield.partnersonline.com, DNS:greenfi
eld.target.com, DNS:hrocdocrequest.target.com, DNS:iccon.target.com, DNS:india.target.com, DNS:jira
.target.com, DNS:launchpad.partnersonline.com, DNS:launchpad.target.com, DNS:m.target.com, DNS:mark
etinghub.target.com, DNS:mercury.partnersonline.com, DNS:mickra.target.com, DNS:mickradashboard.tar
get.com, DNS:mvs.partnersonline.com, DNS:mytime.target.com, DNS:nic.target, DNS:openhouse.target.co
m, DNS:opensource.target.com, DNS:osmosis.partnersonline.com, DNS:partnersonline.com, DNS:pcn.partn
ersonline.com, DNS:peg.partnersonline.com, DNS:photosubmission.target.com, DNS:pid.partnersonline.c
om, DNS:plus.target.com, DNS:pmworkorderadmin.partnersonline.com, DNS:poladmin.partnersonline.com,
DNS:pop.partnersonline.com, DNS:qr.target.com, DNS:r2d2.target.com, DNS:rdmplus.target.com, DNS:rec
ognize.target.com, DNS:reedcard.target.com, DNS:rik.roundel.com, DNS:roundel.com, DNS:rubix.partners
online.com, DNS:rubix.target.com, DNS:security.target.com, DNS:servicetech.target.com, DNS:sm.partn
ersonline.com, DNS:spark.partnersonline.com, DNS:spark.target.com, DNS:stylehub.target.com, DNS:syn
ergy.partnersonline.com, DNS:target.com, DNS:targetmedianetwork.target.com, DNS:tepagent.target.com
, DNS:tgt-files.target.com, DNS:tgtdriver.partnersonline.com, DNS:tiam.target.com, DNS:tiiam.target
.com, DNS:tvi.partnersonline.com, DNS:viewpoint.target.com, DNS:weeklyad.target.com, DNS:www.partne
rsonline.com, DNS:www.target.com
| Not valid before: 2025-09-04T16:25:19
|_Not valid after:  2026-10-06T16:25:18
|_ssl-date: TLS randomness does not represent time
| fingerprint-strings:
|   FourOhFourRequest:
|     HTTP/1.1 421 Misdirected Request
|     Connection: close
|     Content-Length: 291
|     content-type: text/plain; charset=utf-8
|     x-served-by: cache-nyc-kteb1890047
|     Requested host does not match any Subject Alternative Names (SANs) on TLS certificate [b0d6d4
ed8c4bd7880c70cd76fdd4b3af4b2949084f95da4065e6df935b303549] in use with this connection.
|     Visit https://www.fastly.com/documentation/guides/concepts/errors/#routing-errors for more in
formation.
|   GetRequest:
|     HTTP/1.1 421 Misdirected Request
|     Connection: close
|     Content-Length: 291
|     content-type: text/plain; charset=utf-8
|     x-served-by: cache-nyc-kteb1890049
|     Requested host does not match any Subject Alternative Names (SANs) on TLS certificate [b0d6d4
ed8c4bd7880c70cd76fdd4b3af4b2949084f95da4065e6df935b303549] in use with this connection.
|     Visit https://www.fastly.com/documentation/guides/concepts/errors/#routing-errors for more in
formation.
|   HTTPOptions:
|     HTTP/1.1 421 Misdirected Request
|     Connection: close
|     Content-Length: 291
|     content-type: text/plain; charset=utf-8
|     x-served-by: cache-nyc-kteb1890073
|     Requested host does not match any Subject Alternative Names (SANs) on TLS certificate [b0d6d4
ed8c4bd7880c70cd76fdd4b3af4b2949084f95da4065e6df935b303549] in use with this connection.
|     Visit https://www.fastly.com/documentation/guides/concepts/errors/#routing-errors for more in
formation.
|   RTSPRequest:
|     HTTP/1.1 400 Bad Request
|     Connection: close
|     Content-Length: 11
|     content-type: text/plain; charset=utf-8
|     x-served-by: cache-nyc-kteb1890060
|     Request
|   tor-versions:
|     HTTP/1.1 400 Bad Request
|     Connection: close
|     Content-Length: 11
|     content-type: text/plain; charset=utf-8
|     x-served-by: cache-nyc-kteb1890086
|_    Request
|_http-title: Site doesn't have a title.
| tls-alpn:
|   h3
|   h2
|   http/1.1
|_  http/1.0
1 service unrecognized despite returning data. If you know the service/version, please submit the f
ollowing fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port443-TCP:V=7.98%T=SSL%I=7%D=2/16%Time=69933E52%P=x86_64-apple-darwin
SF:21.6.0%r(GetRequest,1BC,"HTTP/1\.1\x20421\x20Misdirected\x20Request\r\n
SF:Connection:\x20close\r\nContent-Length:\x20291\r\ncontent-type:\x20text
SF:/plain;\x20charset=utf-8\r\nx-served-by:\x20cache-nyc-kteb1890049\r\n\r
SF:\nRequested\x20host\x20does\x20not\x20match\x20any\x20Subject\x20Altern
```

(Figure 10: Comprehensive Scan Output)

Purpose

This scan was done to identify services running on open ports and detect their versions. It also runs default Nmap scripts to gather more information.

Results

Port 80 is open. Service detected: http proxy Varnish. Port 443 is open. Service detected: ssl https. The scan returned several HTTP responses such as 400 Bad Request and 421 Misdirected Request. It also shows many DNS names related to target.com.

Analysis

The server is using Varnish as a proxy for web traffic. The HTTPS service is active and uses SSL. The HTTP errors appear because the scan connects without proper headers. This is normal

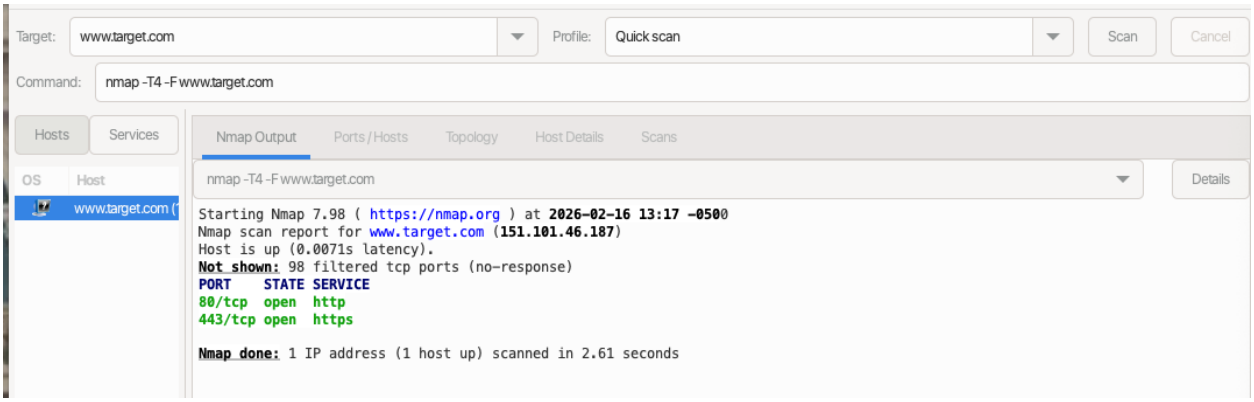
behavior during version detection. The scan confirms that web services are running on ports 80 and 443.

Zenmap Scanning Section

Zenmap

Quick Scan

Profile Used – Quick Scan




(Figure 11: Zenmap Quick Scan Output)

Command generated by Zenmap

nmap -T4 -F www.target.com

Purpose

This scan was performed using the Zenmap graphical interface to quickly identify open ports on the target system.

Results

The scan detected two open ports:

Port 80 for HTTP.  Port 443 for HTTPS

Analysis

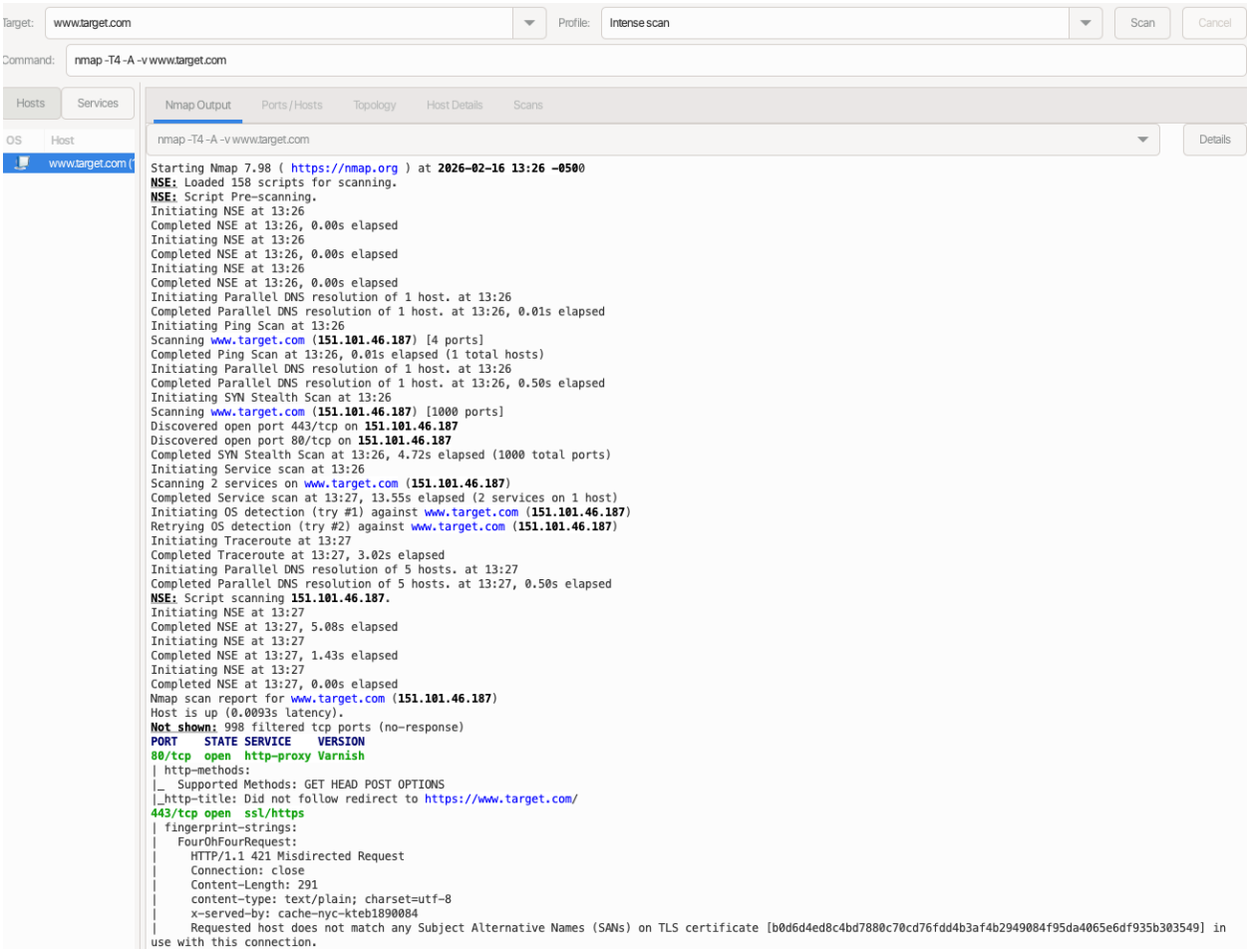
The results match the terminal scan. The server allows web services and blocks other ports.

Intense Scan

Profile Used – Intense Scan

Command generated by Zenmap

nmap -T4 -A -v www.target.com




(Figure 12: Zenmap Intense Scan Output)

Target 2- www.macys.com

Organization: Macys Inc.

Scanning Tool: Nmap 7.98

Operating System Used: macOS (Terminal)

Step 1- Host Discovery Scan

Command Used

nmap -sP www.macys.com

Screen Shot

```
rajulama@MacBookPro ~ % nmap -sP www.macys.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 14:53 -0500
Nmap scan report for www.macys.com (23.56.162.253)
Host is up (0.0058s latency).
rDNS record for 23.56.162.253: a23-56-162-253.deploy.static.akamaitechnologies.com
Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
rajulama@MacBookPro ~ %
```

(Figure 13: Macys Host Discovery Scan Output)

Purpose

This scan checks whether the target host is active and reachable before performing further scans.

Results

The scan confirmed that www.macys.com is up. The host responded successfully and an IP address was resolved.

Analysis

The response confirms that the target system is reachable and ready for further scanning.

Step 2- TCP Ping Scan

Command Used

nmap -sP -PT80 www.macys.com

Screen Shot

```
rajulama@MacBookPro ~ % nmap -sP -PT80 www.macys.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 14:59 -0500
Nmap scan report for www.macys.com (23.197.180.190)
Host is up (0.013s latency).
rDNS record for 23.197.180.190: a23-197-180-190.deploy.static.akamaitechnologies.com
Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
```

(Figure 14: Macys TCP Ping Scan Output)

Purpose

This scan checks if the host is alive by sending a TCP packet to port 80 instead of using normal ICMP ping.

Results

The scan confirmed that the host is up and responding to TCP traffic.

Analysis

The response shows that web traffic to port 80 is allowed. This confirms that the website is reachable through HTTP services.

Step 3- TCP Connect Scan

Command Used

nmap -T4 -sT www.macys.com

Screen Shot

```
rajulama@MacBookPro ~ % nmap -T4 -sT www.macys.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 15:04 -0500
Nmap scan report for www.macys.com (23.222.124.232)
Host is up (0.0065s latency).
rDNS record for 23.222.124.232: a23-222-124-232.deploy.static.akamaitechnologies.com
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 43.18 seconds
```

(Figure 15: Macys TCP Connect Scan Output)

Purpose

The TCP connect scan was performed to identify open TCP ports on the target system.

The -T4 option increases the scan speed.

Results

The scan identified two open ports:

Port 80 (HTTP)

Port 443 (HTTPS)

All other scanned ports were filtered.

Analysis

The presence of ports 80 and 443 confirms that the website provides web services. The Filtered ports indicate that a firewall is blocking access to other services.

Step 4- SYN Stealth Scan

Command Used

sudo nmap -sS -Pn www.macys.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sS -Pn www.macys.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 17:52 -0500
Nmap scan report for www.macys.com (23.222.124.232)
Host is up (0.0090s latency).
rDNS record for 23.222.124.232: a23-222-124-232.deploy.static.akamaitechnologies.com
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 6.85 seconds
```

(Figure 16: Macys SYN Scan Output)

Purpose

This scan checks open ports using a SYN scan. It does not complete the full TCP handshake. It is used to perform a quieter scan.

Results

The scan showed that port 80 and port 443 are open. All other ports were filtered.

Analysis

The results confirm that the website exposes web services only. The filtered ports indicate that firewall protection is active.

Step 5- UDP Scan

Command Used

sudo nmap -sU www.macys.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sU www.macys.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 17:59 -0500
Nmap scan report for www.macys.com (23.62.10.224)
Host is up (0.0055s latency).
rDNS record for 23.62.10.224: a23-62-10-224.deploy.static.akamaitechnologies.com
All 1000 scanned ports on www.macys.com (23.62.10.224) are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 22.68 seconds
```

(Figure 17: Macys UDP Scan Output)

Purpose

This scan checks for open UDP ports on the target system.

Results

The scan showed that all 1000 UDP ports are in ignored or open filtered ststes. No specific UDP services responded.

Analysis

No UDP services were clearly identified. This suggests that the firewall is filtering UDP traffic. The server allows only necessary services.

Step 6- FIN Stealth Scan

Command Used

sudo nmap -sF www.macys.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sF www.macys.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 18:04 -0500
Nmap scan report for www.macys.com (23.56.162.253)
Host is up (0.0066s latency).
rDNS record for 23.56.162.253: a23-56-162-253.deploy.static.akamaitechnologies.com
All 1000 scanned ports on www.macys.com (23.56.162.253) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.36 seconds
```

(Figure 18: Macys FIN Scan Output)

Purpose

This scan sends TCP packets with the FIN flag set. It is used to test firewall filtering behavior.

Results

The scan showed that all 1000 TCP ports are in ignored states. No open ports were detected.

Analysis

The server did not respond to the FIN scan packets. This indicates that a firewall is filtering unusual scan traffic and blocking this type of stealth scan.

Step 7- Xmas Scan

Command Used

sudo nmap -sX www.macys.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sX www.macys.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 18:09 -0500
Nmap scan report for www.macys.com (23.197.44.167)
Host is up (0.0068s latency).
rDNS record for 23.197.44.167: a23-197-44-167.deploy.static.akamaitechnologies.com
All 1000 scanned ports on www.macys.com (23.197.44.167) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 22.06 seconds
```

(Figure 19: Macys Xmas Scan Output)

Purpose

This scan sends TCP packets with FIN, URG, and PUSH flags set. It is used to test firewall behavior.

Results

The scan showed that all 1000 TCP ports are in ignored states. No open ports were detected.

Analysis

The target system did not respond to the Xmas Scan packets. This indicates that firewall protection is active and blocking unusual scan techniques.

Step 8- Null Scan

Command Used

sudo nmap -sN www.macys.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sN www.macys.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 18:14 -0500
Nmap scan report for www.macys.com (23.34.124.247)
Host is up (0.011s latency).
rDNS record for 23.34.124.247: a23-34-124-247.deploy.static.akamaitechnologies.com
All 1000 scanned ports on www.macys.com (23.34.124.247) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 22.24 seconds
```

(Figure 20: Macys Null Scan Output)

Purpose

This scan sends TCP packets with no flag set. It is used to test firewall filtering rules and detect open ports without completing a normal connection.

Results

The scan showed that all 1000 TCP ports are in ignored states. No open ports were detected.

Analysis

The target system did not respond to the Null scan packets. This indicates that a firewall is blocking abnormal or suspicious traffic. The server is properly protected against this stealth method.

Step 9- OS Detection

Command Used

sudo nmap -sS -O www.macys.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sS -O www.macys.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 18:20 -0500
Nmap scan report for www.macys.com (23.222.124.232)
Host is up (0.0087s latency).
rDNS record for 23.222.124.232: a23-222-124-232.deploy.static.akamaitechnologies.com
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed po
rt
Aggressive OS guesses: Linux 4.15 - 5.19 (94%), Linux 4.19 (94%), Linux 5.0 - 5.14 (94%), OpenWrt 2
1.02 (Linux 5.4) (94%), Linux 6.0 (94%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (94%), Linux 2.6
.39 (93%), Linux 2.6.22 - 2.6.36 (89%), Linux 5.4 - 5.10 (88%), Linux 2.6.32 (88%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.08 seconds
```

(Figure 21: Macys OS Detection Output)

Purpose

This scan attempts to identify the operating system of the target host using TCP/IP fingerprinting techniques.

Results

Port 80 (HTTP) and port 443 (HTTPS) were detected as open.

Nmap provided several possible OS matches including:

- Linux 4.15 – 5.19
- Linux 5.0 – 5.14

- OpenWrt
- MikroTik RouterOS
- Linux 2.6.x

The scan also displayed the message:

“No exact OS matches for host (test conditions non-ideal).”

Analysis

The OS detection did not identify an exact operating system but suggested multiple Linux-based systems. The reverse DNS record indicates the server is behind Akamai, a content delivery network. CDN protection and network filtering make accurate OS fingerprinting difficult, so the exact version cannot be confirmed.

Step 10- Comprehensive Scan

sudo nmap -sC -sV www.macys.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sC -sV www.macys.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 18:29 -0500
Nmap scan report for www.macys.com (23.197.180.190)
Host is up (0.011s latency).
rDNS record for 23.197.180.190: a23-197-180-190.deploy.static.akamaitechnologies.com
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE  VERSION
80/tcp  open  http     AkamaiGHost (Akamai's HTTP Acceleration/Mirror service)
|_http-server-header: AkamaiNetStorage
|_http-title: Access Denied
443/tcp open  ssl/http AkamaiGHost (Akamai's HTTP Acceleration/Mirror service)
|_http-server-header: AkamaiNetStorage
|_ssl-date: TLS randomness does not represent time
| ssl-cert: Subject: commonName=www.macys.com/organizationName=Macy's Systems and Technology, Inc./
stateOrProvinceName=Georgia/countryName=US
| Subject Alternative Name: DNS:www.macys.com, DNS:assets.macys.com, DNS:assets.macysassets.com, DN
S:imp-prod.macys.com, DNS:m.macys.com, DNS:macys.com, DNS:secure-m.macys.com, DNS:secure-netstorage
.macys.com, DNS:sell-qa.macys.com, DNS:sell.macys.com, DNS:slimages.macys.com, DNS:slimages.macysas
sets.com, DNS:wc.assets.macys.com, DNS:wc.slimages.macys.com, DNS:www.assets.macys.com, DNS:www.ass
ets.macysassets.com, DNS:www.slimages.macys.com, DNS:www.slimages.macysassets.com, DNS:www.wc.asset
s.macys.com, DNS:www.wc.slimages.macys.com, DNS:www.www1.macys.com, DNS:www1.macys.com
| Not valid before: 2025-07-16T00:00:00
|_Not valid after:  2026-07-16T23:59:59
|_http-title: Access Denied

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.49 seconds
```

(Figure 22: Macys Comprehensive Scan Output)

Purpose

This scan identifies service versions running on open ports and executes default Nmap scripts to gather additional information.

Results

Port 80 is open and running AkamaiGHost (Akamai HTTP Acceleration/Mirror service). Port 443 is open and running ssl/http AkamaiGHost.

The scan returned SSL certificate details showing the organization name Macy’s System and Technology, Inc. It also displayed multiple DNS entries related to macys.com.

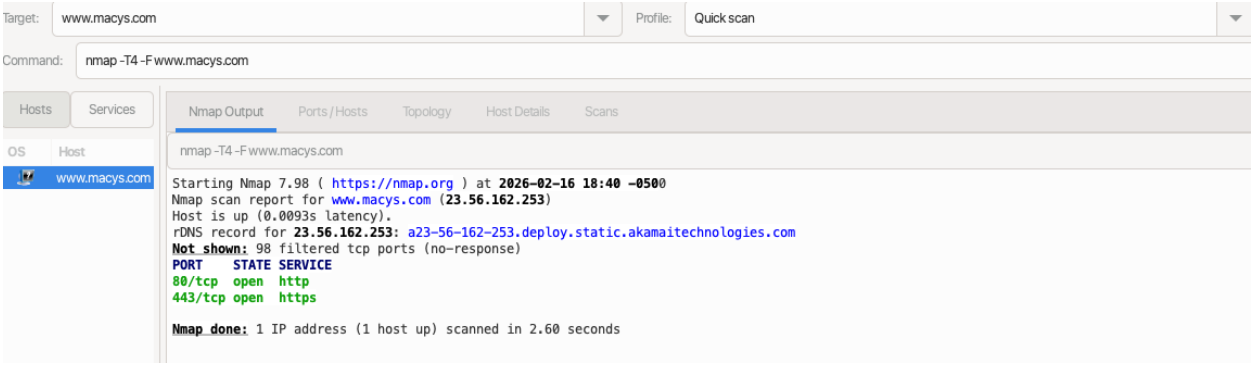
Analysis

Ther results show that Macys is using Akamai as a content delivery and acceleration service. Bot HTTP and HTTPS traffic are handled through Akamai infrastructure. The SSL certificate confirms the organization identity. This indicates enterprise-level web infrastructure and security implementation.

Zenmap Section

Quick Scan

Profile Used: Quick Scan




(Figure 23: Macys Zenmap Quick Scan Output)

Command generated by Zenmap

Nmap -T4 -F www.macys.com

Purpose

This scan was performed using the Zenmap graphical interface to quickly ifdentify open ports on the target system.

Results

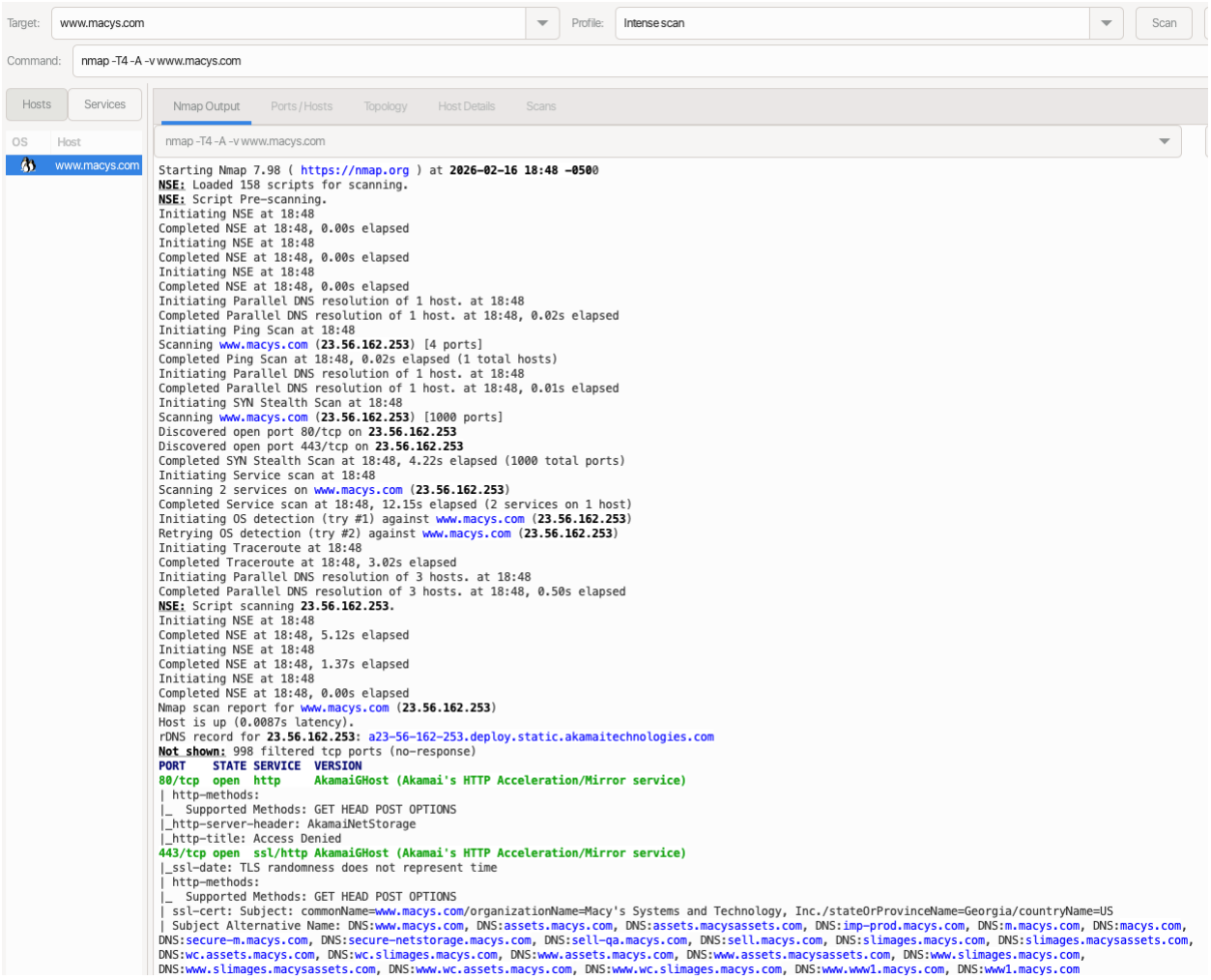
The scan detected two open ports. That are Port 80 (HTTP) and Port 443 (HTTPS).

Analysis

The results match the terminal scan. The server allows only web services and blocks other ports. This confirms consistency between command-line Nmap and Zenmap results.

Intense Scan

Profile Used- Intense Scan




(Figure 24: Macys Zenmap Intense Scan Output)

Command generated by Zenmap

Nmap -T4-A-v www.macys.com

Purpose

This scan performs aggressive scaning including OS detection, version detection, script, and tracerote using the Zenmap graphical interface.

Results

The scan detected:

- Port 80 open running AkamaiGHost
- Port 443 open running ssl/http AkamaiGhost
- OS detection attempts with no exact match
- Multiple DNS and SSL certificate details

Analysis

The ruslts match the terminal comprehensive scan. The presence of Akamai confirms that Macys uses a content delivery network for web traffic acceleration and security. Zenmap produces consistent results with the command-line Nmap scans.

Two Small Sites

Target 3- www.joespizzanyc.com

Organization: Joe's Pizza

Scanning Tool: Nmap 7.98

Operating System Used: macOS (Terminal)

Step 1- Host Discovery Scan

Commnad Used

nmap -sn www.joespizzanyc.com

Screen Shot

```
rajulama@MacBookPro ~ % nmap -sn www.joespizzanyc.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 08:32 -0500
Nmap scan report for www.joespizzanyc.com (198.49.23.144)
Host is up (0.0081s latency).
Other addresses for www.joespizzanyc.com (not scanned): 198.185.159.144 198.49.23.145 198.185.159.1
45
Nmap done: 1 IP address (1 host up) scanned in 0.85 seconds
```

(Figure 25: Joe's Pizza Host Discovery Scan Output)

Purpose

This scan checks whether the target host is active and reachable without scanning ports.

Results

The scan reported that 1 IP address is up. The host responded successfully.

Analysis

The result confirms that the website is reachable. Since the host is active, further port scanning can be performed.

Step 2- TCP Conect Scan

Command Used

nmap -T4 -sT www.joespizzanyc.com

Screen Shot

```
rajulama@MacBookPro ~ % nmap -T4 -sT www.joespizzanyc.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 08:45 -0500
Nmap scan report for www.joespizzanyc.com (198.185.159.144)
Host is up (0.0088s latency).
Other addresses for www.joespizzanyc.com (not scanned): 198.49.23.145 198.185.159.145 198.49.23.144
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE  SERVICE
43/tcp   closed whois
80/tcp   open   http
443/tcp  open   https
2030/tcp closed device2

Nmap done: 1 IP address (1 host up) scanned in 43.53 seconds
rajulama@MacBookPro ~ %
```

(Figure 26: Joe's Pizza TCP Connect Scan Output)

Purpose

The TCP Connect scan was performed to identify open TCP ports on the target system. The -T4 option increases scan speed.

Results

The scan identified two open ports:

Port 80 (HTTP)

Port 443 (HTTPS)

Several other ports were closed.

Analysis

The presence of ports 80 and 443 indicates that the website provides web services. Closed ports show that no additional services are publicly exposed. This suggests basic firewall protection.

Step 3- SYN Stealth Scan

Command Used

sudo nmap -sS -Pn www.joespizzanyc.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sS -Pn www.joespizzanyc.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 08:50 -0500
Nmap scan report for www.joespizzanyc.com (198.49.23.145)
Host is up (0.0085s latency).
Other addresses for www.joespizzanyc.com (not scanned): 198.185.159.145 198.49.23.144 198.185.159.1
44
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE  SERVICE
43/tcp   closed whois
80/tcp   open   http
443/tcp  open   https
2030/tcp closed device2

Nmap done: 1 IP address (1 host up) scanned in 5.83 seconds
```

(Figure 27: Joe's Pizza SYN Stealth Scan Output)

Purpose

This scan checks open ports using a SYN scan without completing the full TCP connection. It is commonly used as a stealth scanning method.

Results

The scan detected port 80 and port 443 as open. The host is up and responding.

Analysis

The results match the TCP Connect scan. The website exposes only web services. This indicates that the server allows HTTP and HTTPS traffic while other services are restricted.

Step 4- UDP Scan

Command Used

sudo nmap -sU www.joespizzanyc.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sU www.joespizzanyc.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 08:54 -0500
Nmap scan report for www.joespizzanyc.com (198.185.159.145)
Host is up (0.0083s latency).
Other addresses for www.joespizzanyc.com (not scanned): 198.49.23.144 198.185.159.144 198.49.23.145
All 1000 scanned ports on www.joespizzanyc.com (198.185.159.145) are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 22.67 seconds
```

(Figure 28: Joe's Pizza UDP Scan Output)

Purpose

This scan checks for open UDP ports on the target system.

Results

The scan showed that all 1000 UDP ports are in ignored states. No open UDP services were detected.

Analysis

The server did not respond to UDP scan requests. This suggests that UDP traffic is filtered or blocked. Only necessary web services appear to be exposed.

Step 5- FIN Stealth Scan

Comand Used

sudo nmap -sF www.joespizzanyc.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sF www.joespizzanyc.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 08:57 -0500
Nmap scan report for www.joespizzanyc.com (198.185.159.145)
Host is up (0.0078s latency).
Other addresses for www.joespizzanyc.com (not scanned): 198.49.23.144 198.185.159.144 198.49.23.145
All 1000 scanned ports on www.joespizzanyc.com (198.185.159.145) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 22.45 seconds
```

(Figure 29: Joe’s Pizza FIN Stealth Scan Output)

Purpose

This scan sends TCP packets with the FIN flag set to test firewall filtering behavior without completing a full connection.

Results

The scan showed that all 1000 TCP ports are in ignored states. No open ports were detected.

Analysis

The server did not respond to the FIN scan packets. This indicates that firewall filtering is active and blocking abnormal scan traffic. The website appears to allow only standard web traffic.

Step 6- Xmas Scan

Command Used

sudo nmap -sX www.joespizzanyc.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sX www.joespizzanyc.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 09:01 -0500
Nmap scan report for www.joespizzanyc.com (198.49.23.145)
Host is up (0.0095s latency).
Other addresses for www.joespizzanyc.com (not scanned): 198.185.159.145 198.49.23.144 198.185.159.1
44
All 1000 scanned ports on www.joespizzanyc.com (198.49.23.145) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 5.00 seconds
```

(Figure 30: Joe's Pizza Xmas Scan Output)

Purpose

This scan sends TCP packets with FIN, URG, and PUSH flags set. It is used to test firewall filtering behavior.

Results

The scan showed that all 1000 TCP ports are in ignored states. No open ports were detected.

Analysis

The server did not respond to the Xmas scan packets. This indicates that firewall protection is filtering unusual scan traffic. Only standard web services are accessible.

Step 7- Null Scan

Command Used

sudo nmap -sN www.joespizzanyc.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sN www.joespizzanyc.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 09:04 -0500
Nmap scan report for www.joespizzanyc.com (198.49.23.144)
Host is up (0.0082s latency).
Other addresses for www.joespizzanyc.com (not scanned): 198.185.159.144 198.49.23.145 198.185.159.1
45
All 1000 scanned ports on www.joespizzanyc.com (198.49.23.144) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.95 seconds
```

(Figure 31: Joe's Pizza Null Scan Output)

Purpose

This scan sends TCP packets with no flags set to test firewall filtering rules.

Results

The scan showed that all 1000 TCP ports are in open|filtered or ignored states. No open ports were detected.

Analysis

The server did not respond to the Null scan packets. This indicates that firewall protection is active and filtering abnormal scan traffic. The website appears to allow only standard web services.

Step 8- OS Detection

Command Used

sudo nmap -sS -O www.joespizzanyc.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sS -O www.joespizzanyc.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 09:10 -0500
Nmap scan report for www.joespizzanyc.com (198.185.159.144)
Host is up (0.017s latency).
Other addresses for www.joespizzanyc.com (not scanned): 198.49.23.145 198.185.159.145 198.49.23.144
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE  SERVICE
43/tcp   closed whois
80/tcp   open   http
443/tcp  open   https
2030/tcp closed device2
Device type: storage-misc|general purpose
Running (JUST GUESSING): HP embedded (85%), FreeBSD 12.X (85%)
OS CPE: cpe:/h:hp:p2000_g3 cpe:/o:freebsd:freebsd:12.0
Aggressive OS guesses: HP P2000 G3 NAS device (85%), FreeBSD 12.0-RELEASE (85%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.46 seconds
```

(Figure 32: Joe’s Pizza OS Detection Scan Output)

Purpose

This scan attempts to identify the operating system of the target using TCP/IP fingerprinting techniques.

Results

Port 80 and port 443 were detected as open.

Nmap identified the device type as storage-misc|general purpose.

The OS guesses included:

- HP embedded device
- FreeBSD 12.x
- HP P2000 G3 NAS device
- FreeBSD 12.0-RELEASE

The scan also displayed the message:"No exact OS matches for host (test conditions non-ideal)."

Analysis

The OS detection suggests a FreeBSD-based or embedded storage system, but no exact match was confirmed. The lack of a precise result indicates filtering or limited fingerprint data. The server may be hosted on a managed hosting platform rather than a standard Linux web server.

Step 9- Comprehensive Scan

Command Used

sudo nmap -sC -sV www.joespizzanyc.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sC -sV www.joespizzanyc.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 09:16 -0500
Nmap scan report for www.joespizzanyc.com (198.185.159.145)
Host is up (0.0084s latency).
Other addresses for www.joespizzanyc.com (not scanned): 198.49.23.144 198.185.159.144 198.49.23.145
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE  SERVICE  VERSION
43/tcp   closed whois
80/tcp   open   http     Golang net/http server
|_http-server-header: Squarespace
|_http-title: Did not follow redirect to https://www.joespizzanyc.com/
| fingerprint-strings:
|   GetRequest:
|     HTTP/1.0 403 Forbidden
|     Connection: close
|     Server: Squarespace
|     X-Contextid: autW0Vwt/Jbbe933I
|     X-Sqsp-Edge: true
|     Date: Tue, 17 Feb 2026 14:16:29 GMT
|     Content-Length: 1434
|     Content-Type: text/html; charset=utf-8
|     <!DOCTYPE html>
|     <head>
```

(Figure 33: Joe's Pizza Comprehensive Scan Output)

Purpose

This scan performs service version detection and executes default Nmap scripts to gather detailed information about open services.

Results

Port 80 is open and running a Golang net/http server. The HTTP server header indicates Squarespace hosting.

Port 443 is open and running HTTPS service.

Analysis

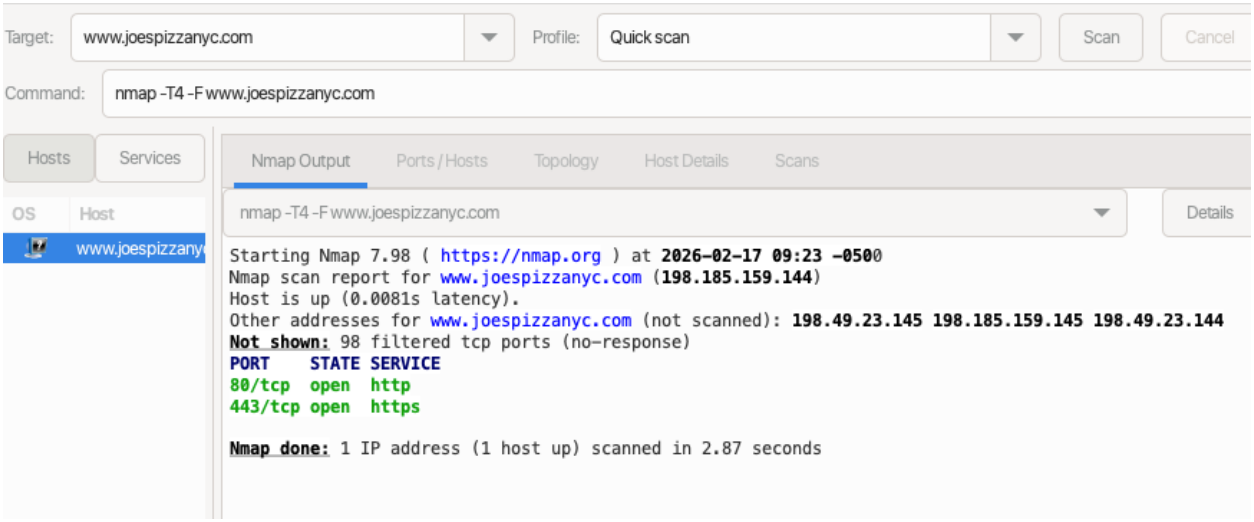
The results show that the website is hosted on Squarespace infrastructure. The use of a Golang HTTP server and HTTPS service indicates a managed hosting platform. Only standard web services are exposed.

Zenmap Scanning Section

Zenmap

Quick Scan

Profile Used – Quick Scan




(Figure 34: Joe's Pizza Zenmap Quick Scan Output)

Command generated by Zenmap

nmap -T4 -F www.joespizzanyc.com

Purpose

This scan was performed using the Zenmap graphical interface to quickly identify open ports on the target system.

Results

The scan detected two open ports:

Port 80 (HTTP)

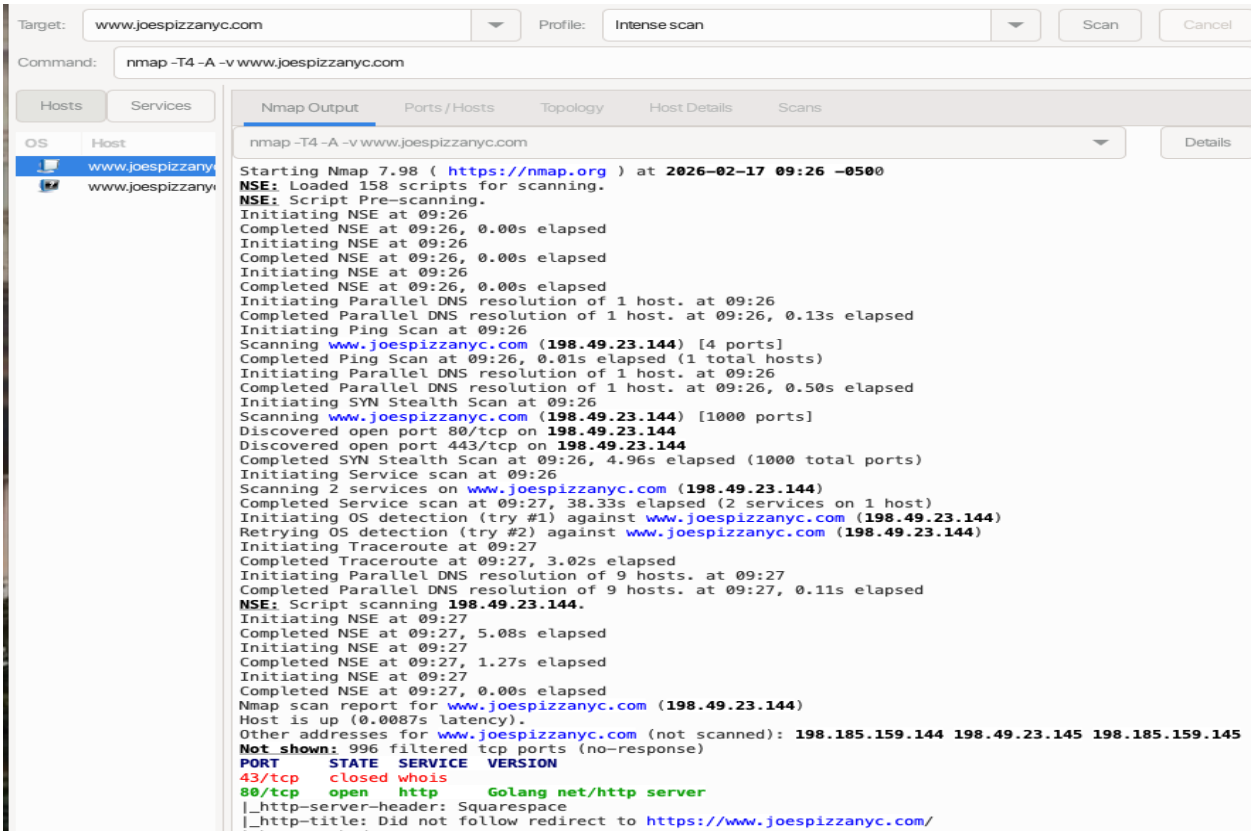
Port 443 (HTTPS)

Analysis

The results are consistent with the command-line scans. The server exposes only web services while other ports remain closed or filtered.

Intense scan

Profile Used – Intense Scan




(Figure 35: Joe's Pizza Zenmap Intense Scan Output)

Command generated by Zenmap

nmap -T4 -A -v www.joespizzanyc.com

Purpose

This scan performs aggressive detection including OS detection, version detection, script scanning, and traceroute using Zenmap.

Results

Port 80 is open running Golang net/http server. Port 443 is open running HTTPS service. The server header indicates Squarespace hosting.

Analysis

The Intense Scan confirms earlier results. The website is hosted on a managed Squarespace platform and only standard web services are exposed. No additional services were detected.

Target 4 – www.bluestockings.com

Organization: Bluestockings Cooperative

Scanning Tool: Nmap 7.98

Operating System Used: macOS (Terminal)

Step 1- Host Discovery Scan

Commnad Used

nmap -sn www.bluestockings.com

Screen Shot

```
rajulama@MacBookPro ~ % nmap -sn www.bluestockings.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 09:34 -0500
Nmap scan report for www.bluestockings.com (35.247.53.102)
Host is up (0.067s latency).
rDNS record for 35.247.53.102: 102.53.247.35.bc.googleusercontent.com
Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
```

(Figure 36: Bluestockings Host Discovery Scan Output)

Purpose

To determine whether the target host is reachable before performing port scanning.

Results

The scan reported 1 IP address is up.

Analysis

The host is active and reachable, so further port scanning can be performed.

Step 2- TCP Connect

Command Used

nmap -T4 -sT www.bluestockings.com

Screen Shot

```
rajulama@MacBookPro ~ % nmap -T4 -sT www.bluestockings.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 09:37 -0500
Nmap scan report for www.bluestockings.com (35.247.53.102)
Host is up (0.070s latency).
rDNS record for 35.247.53.102: 102.53.247.35.bc.googleusercontent.com
Not shown: 923 filtered tcp ports (no-response), 72 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
10001/tcp open  scp-config

Nmap done: 1 IP address (1 host up) scanned in 42.07 seconds
```

(Figure 37: Bluestockings TCP Connect Scan Output)

Purpose

To identify open TCP ports and exposed services on the target system.

Results

The scan detected the following open ports:

- 21 (FTP)
- 22 (SSH)
- 80 (HTTP)
- 443 (HTTPS)
- 1001 (Unknown/Custom Service)

72 ports were closed.

Analysis

Unlike previous targets, this server exposes multiple services including FTP and SSH. This indicates a less restricted configuration and potentially a larger attack surface. Administrative services such as SSH being publicly accessible may increase security risk if not properly secured.

Step 3- SYN Stealth Scan

Command Used

sudo nmap -sS -Pn www.bluestockings.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sS -Pn www.bluestockings.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 09:43 -0500
Nmap scan report for www.bluestockings.com (35.247.53.102)
Host is up (0.069s latency).
rDNS record for 35.247.53.102: 102.53.247.35.bc.googleusercontent.com
Not shown: 923 filtered tcp ports (no-response), 72 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
10001/tcp open  scp-config

Nmap done: 1 IP address (1 host up) scanned in 4.64 seconds
```

(Figure 38: Bluestockings SYN Stealth Scan Output)

Purpose

To perform a stealth TCP SYN scan to identify open ports without completing full TCP connections.

Results

The scan confirmed the following open ports:

- 21 (FTP)
- 22 (SSH)
- 80 (HTTP)
- 443 (HTTPS)
- 1001 (Unknown/Custom Service)

Analysis

The SYN scan results match the TCP Connect scan, confirming that these services are actively listening. The presence of FTP and SSH indicates that remote management services are accessible from the internet, which increases the attack surface if not properly secured.

Step 4- UDP Scan

Command Used

sudo nmap -sU -Pn www.bluestockings.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sU --top-ports 20 -Pn www.bluestockings.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 09:53 -0500
Nmap scan report for www.bluestockings.com (35.247.53.102)
Host is up.
rDNS record for 35.247.53.102: 102.53.247.35.bc.googleusercontent.com

PORT      STATE         SERVICE
53/udp    open|filtered domain
67/udp    open|filtered dhcps
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
123/udp   open|filtered ntp
135/udp   open|filtered msrpc
137/udp   open|filtered netbios-ns
138/udp   open|filtered netbios-dgm
139/udp   open|filtered netbios-ssn
161/udp   open|filtered snmp
162/udp   open|filtered snmptrap
445/udp   open|filtered microsoft-ds
500/udp   open|filtered isakmp
514/udp   open|filtered syslog
520/udp   open|filtered route
631/udp   open|filtered ipp
1434/udp  open|filtered ms-sql-m
1900/udp  open|filtered upnp
4500/udp  open|filtered nat-t-ike
49152/udp open|filtered unknown

Nmap done: 1 IP address (1 host up) scanned in 5.82 seconds
```

(Figure 39: Bluestockings UDP Scan Output)

Purpose

To identify open UDP services on the target using the most common UDP ports.

Results

All 20 scanned UDP ports returned the state open|filtered, including:

- 53 (DNS)
- 123 (NTP)
- 161 (SNMP)
- 1900 (UPnP)
- 500 (ISAKMP)
- 4500 (NAT-T)

Analysis

The open/filtered state indicates that the host did not respond to UDP probes. This commonly occurs when a firewall filters UDP traffic. No UDP services were definitively confirmed as open. The server appears to be hosted on Google Cloud infrastructure, as indicated by the reverse DNS record.

Step 5- FIN Stealth Scan

Command Used

sudo nmap -sF -Pn www.bluestockings.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sF -Pn www.bluestockings.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 10:02 -0500
Nmap scan report for www.bluestockings.com (35.247.53.102)
Host is up.
rDNS record for 35.247.53.102: 102.53.247.35.bc.googleusercontent.com
All 1000 scanned ports on www.bluestockings.com (35.247.53.102) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 202.55 seconds
```

(Figure 40: Bluestockings FIN Stealth Scan Output)

Purpose

To perform a stealth FIN scan to identify open ports by sending TCP packets with the FIN flag set. This technique attempts to bypass basic firewall logging.

Results

All 1000 scanned TCP ports were reported as open/filtered (no response).

Analysis

The lack of response indicates that the target system or its firewall is filtering FIN probe packets. Modern firewalls commonly drop these packets instead of responding, which prevents accurate port state determination. This suggests that the server is protected by firewall rules that restrict abnormal TCP flag scans.

Step 6- NULL Scan

Command Used

sudo nmap -sN -Pn www.bluestockings.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sN -Pn www.bluestockings.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 10:10 -0500
Nmap scan report for www.bluestockings.com (35.247.53.102)
Host is up.
rDNS record for 35.247.53.102: 102.53.247.35.bc.googleusercontent.com
All 1000 scanned ports on www.bluestockings.com (35.247.53.102) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 202.47 seconds
```

(Figure 41: Bluestockings NULL ScanScan Output)

Purpose

To perform a NULL scan by sending TCP packets with no flags set in order to detect open ports while attempting to evade simple firewall detection.

Results

All 1000 scanned TCP ports were reported as open/filtered (no response).

Analysis

The absence of responses indicates that the server or its firewall is filtering NULL probe packets. Modern firewall configurations typically drop abnormal TCP flag packets, preventing accurate port state detection. This behavior confirms that the host is protected against stealth scan techniques.

Step 7- Xmas Scan

Command Used

sudo nmap -sX -Pn www.bluestockings.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -sX -Pn www.bluestockings.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 10:17 -0500
Nmap scan report for www.bluestockings.com (35.247.53.102)
Host is up.
rDNS record for 35.247.53.102: 102.53.247.35.bc.googleusercontent.com
All 1000 scanned ports on www.bluestockings.com (35.247.53.102) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 202.45 seconds
```

(Figure 42: Bluestockings Xmas Scan Output)

Purpose

To perform an Xmas scan by sending TCP packets with FIN, PSH, and URG flags set in order to identify open ports while attempting to evade basic intrusion detection systems.

Results

All 1000 scanned TCP ports were reported as open/filtered (no response).

Analysis

The absence of responses indicates that the server or its firewall is dropping abnormal TCP flag packets. Modern firewall configurations commonly block Xmas scans, preventing Nmap from

distinguishing between open and filtered ports. This confirms that the target system is protected against stealth scanning techniques.

Step 8- OS Detection

Command Used

sudo nmap -O -Pn www.bluestockings.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -O -Pn www.bluestockings.com

Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 10:23 -0500
Nmap scan report for www.bluestockings.com (35.247.53.102)
Host is up (0.070s latency).
rDNS record for 35.247.53.102: 102.53.247.35.bc.googleusercontent.com
Not shown: 923 filtered tcp ports (no-response), 72 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
10001/tcp open  scp-config
Aggressive OS guesses: Linux 2.6.32 - 3.13 (93%), Linux 5.0 - 5.14 (93%), MikroTik RouterOS 7.2 - 7
.5 (Linux 5.6.3) (93%), Linux 3.10 - 4.11 (91%), Linux 3.2 - 4.14 (90%), Linux 4.15 (90%), Linux 4.
15 - 5.19 (90%), Linux 4.19 (90%), Linux 2.6.32 - 3.10 (90%), MikroTik RouterOS 6.36 - 6.48 (Linux
3.3.5) (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.17 seconds
```

(Figure 43: Bluestockings OS Detection Scan Output)

Purpose

To identify the operating system running on the target host using OS fingerprinting.

Results

Nmap detected multiple possible Linux-based systems (90–93% accuracy) but did not identify one exact version. Several ports were open (21, 22, 80, 443, 10001).

Analysis

The results indicate the server is likely running Linux. The lack of an exact match is due to firewall filtering and cloud hosting (Google Cloud), which limits precise OS fingerprinting.

Step 9- Comprehensive Scan

Command Used

sudo nmap -A -Pn www.bluestockings.com

Screen Shot

```
rajulama@MacBookPro ~ % sudo nmap -A -Pn www.bluestockings.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-17 10:25 -0500
Nmap scan report for www.bluestockings.com (35.247.53.102)
Host is up (0.070s latency).
rDNS record for 35.247.53.102: 102.53.247.35.bc.googleusercontent.com
Not shown: 923 filtered tcp ports (no-response), 72 closed tcp ports (reset)
PORT      STATE SERVICE         VERSION
21/tcp    open  ftp             ProFTPD
22/tcp    open  ssh             OpenSSH 10.0 (protocol 2.0)
80/tcp    open  http            Apache httpd 2.4.63
|_http-server-header: Apache/2.4.63 (Unix)
|_http-title: Did not follow redirect to https://www.bluestockings.com/
443/tcp   open  ssl/http        Apache httpd 2.4.63
| ssl-cert: Subject: commonName=www.bluestockings.com
| Subject Alternative Name: DNS:www.bluestockings.com
| Not valid before: 2026-01-28T19:05:34
|_Not valid after:  2026-04-28T19:05:33
|_ssl-date: TLS randomness does not represent time
| http-robots.txt: 1 disallowed entry
|_/i/
|_http-server-header: Apache/2.4.63 (Unix)
|_http-title: Bluestockings Cooperative
10001/tcp open  ssl/http-proxy HAProxy http proxy
| ssl-cert: Subject: commonName=www.bluestockings.com
| Subject Alternative Name: DNS:www.bluestockings.com
```

(Figure 44: Bluestockings Comprehensive Scan Output)

Purpose

To perform an in-depth scan including OS detection, service version detection, and default script scanning.

Results

The scan identified open ports 21 (ProFTPD), 22 (OpenSSH 10.0), 80 and 443 (Apache 2.4.63). SSL certificate information for the domain was also detected. Most other ports were filtered.

Analysis

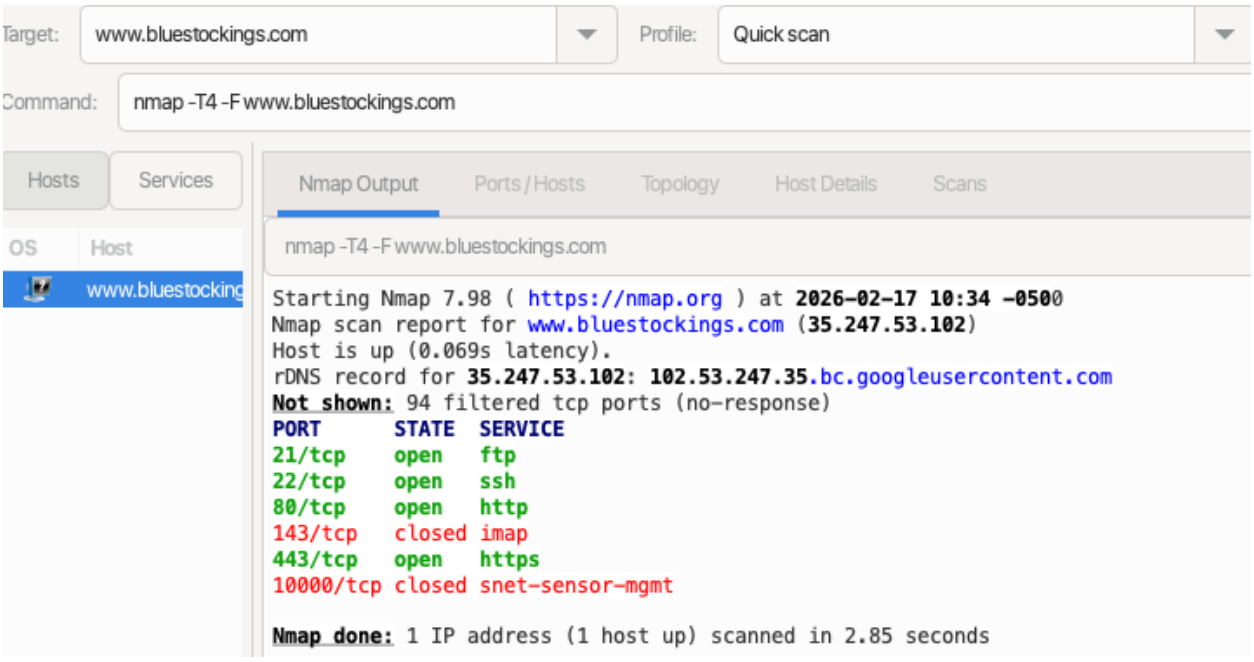
The comprehensive scan confirms the host is running Linux with Apache web services and secure remote access via SSH. The presence of filtered ports indicates firewall protection. Service versions were successfully identified, providing detailed insight into the server configuration.

Zenmap Scanning Section

Zenmap

Quick Scan

Profile Used- Quick Scan




(Figure 45: Bluestockings Zenmap Quick Scan Output)

Command generated by Zenmap

nmap -T4 -F www.bluestockings.com

Purpose

To perform a basic scan using Zenmap's Quick Scan profile.

Results

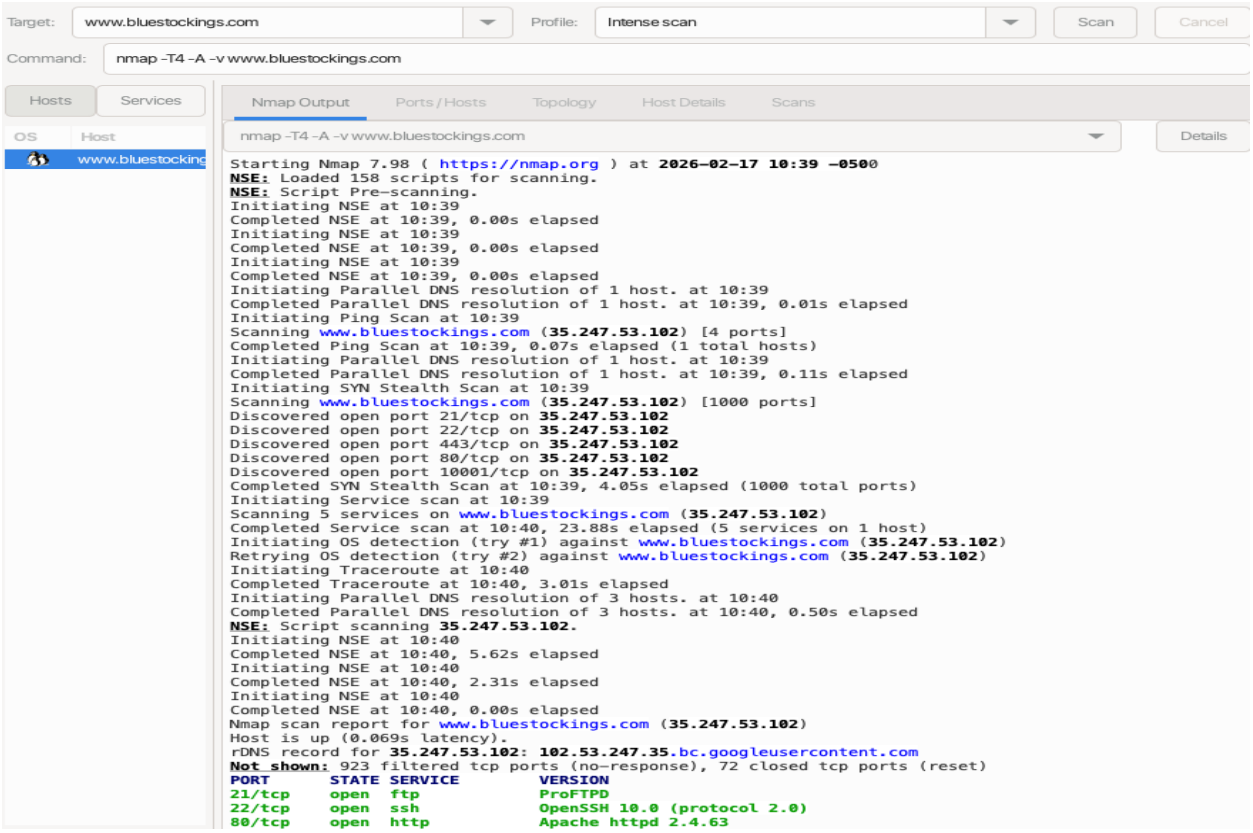
The scan identified open ports 21, 22, 80, and 443.

Analysis

The Quick Scan confirms the presence of web and remote access services. The results match the earlier Nmap command-line scan.

Intense Scan

Profile Used – Intense Scan




(Figure 46: Bluestockings Zenmap Intense Scan Output)

Purpose

To perform a detailed scan including OS detection, version detection, and script scanning.

Results

The scan detected service versions (ProFTPD, OpenSSH 10.0, Apache 2.4.63) and indicated a Linux-based operating system.

Analysis

The Intense Scan provides deeper insight into the server configuration and confirms previous findings from the Nmap comprehensive scan.

Conclusion

This project showed how Nmap can be used to scan public websites and gather information about them. Different types of scans were used to find open ports, running services, service versions, and possible operating systems. The results showed that most large webistes use firewalls and content delivery networks to protect their servers and block unnecessary traffic.

Smaller websites usually allow only basic web services such as HTTP and HTTPS. Overall, this project helped me understand how network scanning works and how servers protect themselves from unwanted access.